Dear Editors:

We would like to submit the enclosed manuscript entitled **“The Vanishing First Rung: A System Dynamics Study of Entry-Level Automation, Senior Time Allocation and the Erosion of the Professional Expertise Pipeline”**, which we wish to be considered for publication in *Systems* for the Special Issue **“Navigating Digital Transformation: Leadership and Decision Making in Today’s Systems.”**

No conflict of interest exists in the submission of this manuscript, and the manuscript is approved by all authors for publication. I would like to declare on behalf of my co-authors that the work described was original research that has not been published previously and is not under consideration for publication elsewhere, in whole or in part. All authors listed have approved the manuscript that is enclosed.

This study asks what happens to a profession when the tasks it has always used to train newcomers can be performed by a machine. Generative artificial intelligence automates precisely the routine, supervised work through which graduates become proficient professionals, so every automation decision a manager takes today silently withdraws a unit of practice whose value would have appeared six to eight years later. We combine two studies. Study 1 surveys 356 knowledge-intensive firms in China’s Yangtze River Delta and estimates, from administrative staffing and personnel records rather than from attitude scales, the behavioural parameters that link automation depth to entry hiring, senior verification time, mentoring intensity and time to independent proficiency. Study 2 embeds those estimates in a five-stock system dynamics model of the entrant pool, the junior and senior workforce and the skill capital of excluded entrants, and subjects it to the full confidence-building protocol for dynamic models—dimensional, equilibrium, extreme-condition, integration-error, structural and behaviour-reproduction tests—before using it to run counterfactual, threshold, decomposition and policy experiments over a thirty-year horizon.

The innovations of this paper are as follows.

**(1) This study supplies the longitudinal, feedback-based account of entry-level AI exposure that the current empirical literature calls for but cannot itself provide.** Recent payroll and vacancy studies observe at most two years of what is a thirty-year adjustment. By making time to independent proficiency an endogenous variable that responds to mentoring, to the learning content of practice and to cohort quality, and by letting the senior workforce feed back into all three, we show that a transitory hiring shock is converted into a persistent capability deficit. The stock of proficient professionals ends 46.3% below the no-automation counterfactual, yet productive capacity first rises 8.7% above it and does not cross back below until year 16.6—so for sixteen years every indicator a firm actually tracks confirms that the decision was correct.

**(2) This study relocates the mechanism, contradicting the account that dominates current commentary.** It is widely asserted that AI erodes professional training because it leaves senior staff with no time to teach. Our survey finds no direct effect of automation depth on mentoring intensity once delivery pressure is controlled, and the simulation shows that automation initially lowers senior utilisation rather than raising it. Structural decomposition attributes 41.2 of the 46.3 percentage-point deficit to task substitution and to the loss of learning content in the entry work that survives, and only 5.7 points to the mentoring squeeze, which engages after year 23 and then produces an abrupt regime shift. We also find that the system's only self-correcting loop is self-defeating: hiring more graduates in response to a visible shortage dilutes the mentoring that converts them, worsening the year-30 outcome by 2.7 percentage points.

**(3) This study identifies a leverage point that differs from the instrument policy debate has settled on, and shows that it is already achievable.** Subsidising entry hiring works but requires 0.306 additional entry hires per additional senior-year created; restoring the learning content of entry work through structured AI-assisted apprenticeship requires only 0.122, is 2.5 times more efficient, and is the only single lever that raises the pipeline's conversion rate above its pre-automation value. Crucially, our survey shows this is not aspirational: the learning penalty of automation is large at the 25th percentile of structured practice and statistically indistinguishable from zero at the 95th percentile, so firms already at the top of the observed distribution avoid the damage entirely, using protocols that require no additional headcount or budget.

We believe that this manuscript fits well within the scope of *Systems* and the Special Issue "Navigating Digital Transformation: Leadership and Decision Making in Today's Systems". The paper is squarely about leadership and decision making: its object of study is the managerial decision of how deeply to automate entry-level work and how to allocate the finite time of senior professionals between delivery, verification of machine output and mentoring. It shows that a sequence of locally rational decisions of this kind produces a systemically costly outcome because the accumulation being depleted lies outside the horizon of the measures that govern the decision, and it identifies which leadership levers actually move the system. The methodology is explicitly systemic—stocks, flows, feedback loops, delays, thresholds and leverage points—which we hope makes the paper a natural fit for the journal's readership, and it complements the survey-based and econometric contributions that the Special Issue has attracted by supplying the longitudinal, feedback-based perspective that those designs cannot provide.

We deeply appreciate your consideration of our manuscript and look forward to receiving comments from the reviewers. Correspondence should be directed to Xinyu Cai at the following address:

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Thanks very much for your attention to our paper.

Very sincerely yours,

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